# Johannes Schwab

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Google Scholar

## Research interests

Machine learning applied to finance, with a focus on asset pricing: high-dimensional tests of asset pricing models, the generalization of machine-learning models, and language models free of lookahead bias

## Education

**École Polytechnique Fédérale de Lausanne (EPFL),** Lausanne, Switzerland
***September 2023 – present***
**PhD in Finance**
*Advisor: Prof. Semyon Malamud (Swiss Finance Institute, EPFL)*

**École Polytechnique Fédérale de Lausanne (EPFL),** Lausanne, Switzerland
***September 2019 – August 2021***
**Master of Science in Financial Engineering,** GPA: 5.8/6
*Master's thesis: Use of machine learning techniques for portfolio insurance (at Vaudoise Assurances)*

**University of Lausanne (UNIL),** Faculty of Business and Economics (HEC), Lausanne, Switzerland
***September 2016 – June 2019***
**Bachelor of Science (BSc) in Economics,** GPA: 5.6/6

## Publications

**Scaling Point-in-Time Language Models** **NeurIPS 2026**
*with Bryan Kelly, Semyon Malamud and Teng Andrea Xu* arXiv:2607.11889 • NBER WP 35247

- Shows that training decoder-only transformers up to 4B parameters on 1T chronologically filtered tokens largely closes the performance gap between lookahead-free language models and their unconstrained counterparts
  *Presented at:* EPFL Computational Future seminar 2026 (F. Kaplan)
  *To be presented at:* NeurIPS 2026 (poster)

## Working papers

**Infinite Factor Structure in Conditional Asset Pricing** *2026*
*with Mikhail Chernov, Bryan Kelly and Semyon Malamud* SSRN 7596842

- Shows that a conditional model with a few factors is, unconditionally, an infinite-factor model in which infinitely many eigenvalues diverge, so no finite set of principal components spans expected returns

**Training NTK to Generalize with KARE** *2025*
*with Bryan Kelly, Semyon Malamud and Teng Andrea Xu* arXiv:2505.11347 • SFI Research Paper 25-51

- Proposes optimizing the neural tangent kernel explicitly rather than implicitly through gradient descent, yielding kernels that match or outperform the deep networks they are derived from

**A Test of the Efficiency of a Given Portfolio in High Dimensions** *2025*
*with Mikhail Chernov, Bryan Kelly and Semyon Malamud* NBER WP 33565 • SFI Research Paper 25-26

- Extends the Gibbons–Ross–Shanken test to settings where test assets far outnumber the sample, using Random Matrix Theory to correct the bias induced by regularized covariance estimation
  *Presented at:* SoFiE 2025 (Paris), ESSFM 2025 (Gerzensee)

## Teaching

**Teaching Assistant,** EPFL, Master in Financial Engineering
***September 2023 – present***

- Stochastic Calculus
- Machine Learning in Finance

## Honors and awards

- **Best Teaching Assistant, 2026 —** awarded by student vote in the EPFL Master in Financial Engineering, for Stochastic Calculus and Machine Learning in Finance
- **Swiss Life Award, 2021 —** awarded to the student graduating with the highest GPA in the EPFL Master of Science in Financial Engineering

## Industry experience

**Quantitative Analyst,** Taranis – Hedge Fund, Geneva, Switzerland
***March 2022 – May 2023***

- Designed a backtesting environment for option-based strategies on a SQL database of over 1 billion data points, developed a Python library for the statistical analysis of trading signals, and co-designed an automated trading bot

**Asset Management Intern,** Vaudoise Assurances, Lausanne, Switzerland
***March 2021 – December 2021***

- Calibrated an option-based portfolio insurance (OBPI) strategy and built Python tools for option pricing, backtesting and reporting

## Skills

- **Programming** — Python (PyTorch, NumPy, pandas, scikit-learn), MATLAB, SQL, LaTeX
- **High-performance computing** — Multi-node GPU training (128 GPUs at CSCS), MPI, NCCL, SLURM

## Languages

- **French** — Native
- **English** — Fluent (C1)
- **German** — Upper intermediate (B2)

*Last updated: October 20, 2026*